# Your Paper Title Here

Your Name(s)

Date

## Abstract

Write your abstract here.

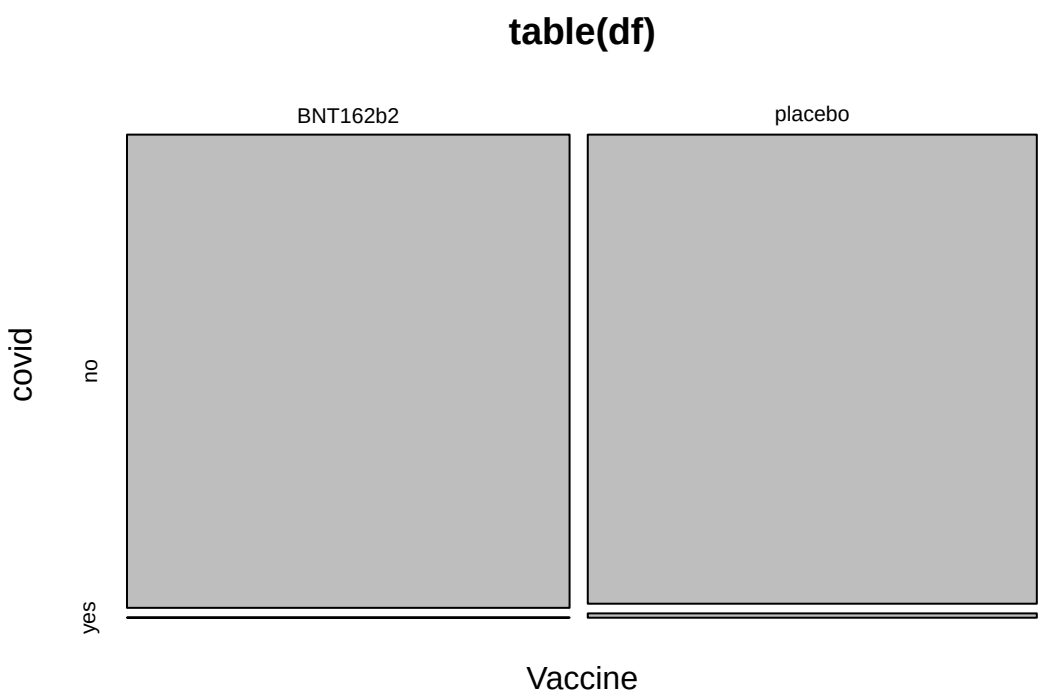
## Keywords

*Bayesian Statistics*, *Likelihood Inference*, *Relative Risk*, *Efficacy*

## Introduction / Background

Provide an introduction with background information.

```
##   Vaccine                 no                yes
##  BNT162b2 0.9995407 (17,411) 0.0004592686   (8)
##   placebo 0.9908335 (17,511) 0.0091665252 (162)
##     Total 1.9903742 (34,922) 0.0096257938 (170)
```




## Statistical Methods

### Model

Here we introduce the statistical model used for inference in this paper. Denote the random variable $T$ as the number in the BNT162b2 vaccine group from the $n = 170$ COVID-19 cases. The

distribution of $T$ can be well approximated by a binomial distribution (see Appendix proof A for proof). Then

$$T \sim Binom(n = 170, \pi)$$

where $\pi$ denotes the probability of an individual being in the vaccine group given they contracted COVID-19.

Let $n_1$, $\pi_v$ and $n_2$, $\pi_p$ be the number of individuals and probability an individual contracts COVID-19 in the vaccine and placebo groups respectively. $\pi = P(\text{vaccine}|\text{COVID-19})$ and by Bayes' theorem can be shown to be equal to

$$\pi = \frac{n_1\pi_v}{n_1\pi_v + n_2\pi_p}$$

We observe $n_1 = 17411$ and $n_2 = 17511$, and since $n_1 \approx n_2$, the randomization is 1:1 and we simplify $\pi$ to

$$\pi = \frac{\pi_v}{\pi_v + \pi_p}$$

We also observe that our event rates $\pi_v$ and $\pi_p$ are small, so we are will use the ratio between the probabilities, or the relative risk $RR = \frac{\pi_v}{\pi_p}$. The estimated relative risk for the COVID-19 vaccine trials is $\widehat{RR} = \frac{\widehat{\pi}_v}{\widehat{\pi}_p} = \frac{0.0004593}{0.0004593} \approx 0.05010$. This means that the risk of contracting COVID-19 in the vaccinated group is around 5% that of the placebo group. This also means that the vaccine reduces the risk of contracting COVID-19 by around 95%, which is the vaccine efficacy.

Denote the parameter of interest $\psi$ to be the vaccine efficacy defined as $1 - RR$:

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{\pi_p - \pi_a}{\pi_p}$$

Writing this in terms of the binomial probability $\pi$, we have our parameter of interest for inference:

$$\psi = \frac{1 - 2\pi}{1 - \pi}$$

Using this single binomial random variable model, we will calculate estimators for $\psi_0$, the true efficacy of the vaccine, and make inference within two frameworks: Likelihood and Bayes. We will

calculate 95% confidence interval estimates for $\psi_0$ and conduct a significance test at the $\alpha = 0.01$ level for testing

$$H_0 : \psi_0 = 0.2 \text{vs. } H_1 : \psi_0 > 0.2$$

Let $T$ be the number from the vaccinated group among the 170 COVID-19 cases. We assume that $T \sim Binom(170, \pi)$ and we observed $t = 8$.

## Likelihood Inference

Detail the likelihood approach.

$\pi = \frac{1-\psi}{2-\psi}$

$L(\pi) = \binom{170}{8}(\pi)^8(1-\pi)^{162}$

$L(\psi) = \binom{170}{8}(\frac{1-\psi}{2-\psi})^8(1-\frac{1-\psi}{2-\psi})^{162} = \binom{170}{8}(\frac{1-\psi}{2-\psi})^8(\frac{2-\psi-1+\psi}{2-\psi})^{162} = \binom{170}{8}(\frac{1-\psi}{2-\psi})^8(\frac{1}{2-\psi})^{162}$

$\ell(\psi) = ln(\binom{170}{8}) + 8(ln(1-\psi) - ln(2-\psi)) + 162(ln(1) - ln(2-\psi))$

$\frac{d}{d\psi}\ell(\psi) = \frac{-8}{1-\psi} + \frac{8}{2-\psi} + \frac{162}{2-\psi} = \frac{170}{2-\psi} - \frac{8}{1-\psi} = 0$

$170 - 170\psi - 16 + 8\psi = 0$

$162\psi = 154$

$\hat{\psi}_0^{mle} = 0.9506173$

We know that the likelihood function for $\pi$ will be $L(\pi) = \binom{170}{8}(\pi)^8(1-\pi)^{162}$, which means the likelihood function of $\psi$ would be $L^*(\psi) = \binom{170}{8}(\frac{1-\psi}{2-\psi})^8(\frac{1}{2-\psi})^{162}$ or $\ell(\psi) = ln(\binom{170}{8}) + 8(ln(1-\psi) - ln(2-\psi)) + 162(ln(1) - ln(2-\psi))$. From this, we can see that $\hat{\psi}_0^{mle} = 0.9506173$, which would be a global maximum.

### Significance Test

We wish to test

$$H_0 : \psi_0 = 0.3, H_1 : \psi_0 > 0.3$$

. We can use the likelihood ratio statistic to calculate a p-value:

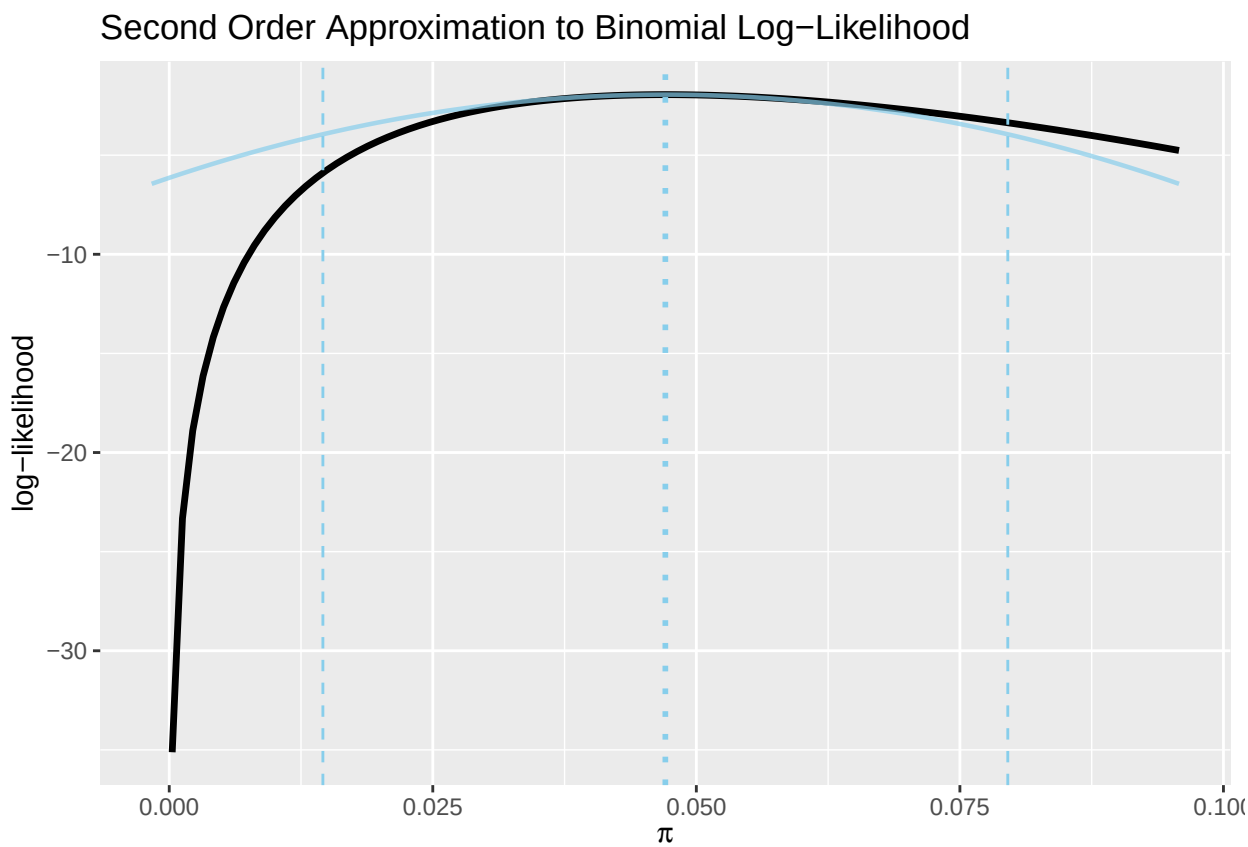
$$\Lambda = \frac{L\left(\widehat{\psi}_0^{mle}\right)}{L\left(\psi_0^{null}\right)}$$

. When the null hypothesis $H_0$ is true, $2 \ln \Lambda \approx \chi_1^2$, when $n \to \infty$. The approximate $\chi_1^2$ distribution on which the likelihood ratio significance test is based relies on the second order approximation to the log-likelihood being accurate. We can verify this by observing a graph of the log likelihood function with its quadratic approximation.

```
loglik_binom <- function(pi, n, x) {
  ifelse((pi <= 0) | (pi >= 1), NA,
    log(choose(n, x)) + x * log(pi) + (n - x) * log(1 - pi)
  )
}

ml <- maxLik2(loglik = loglik_binom, start = 0.5, n = 170, x = 8)

plot(ml) +
  labs(title = "Second Order Approximation to Binomial Log-Likelihood",
       x = expression(pi))
```




Our log-likelihood function doesn't closely resemble its quadratic approximation and isn't symmetric around the MLE, which is a reason to concerned about approximating our likelihood ratio with a $\chi_1^2$ distribution. We can verify our results from the p-value from the $\chi_1^2$ distribution by calculating an empirical p-value.

We know that $L(\psi) = \binom{170}{8}(\frac{1-\psi}{2-\psi})^8(1-\frac{1-\psi}{2-\psi})^{162}$, so we get the following values for $L\left(\widehat{\psi}_0^{mle}\right)$ and $L\left(\psi_0^{null}\right)$:

$$
\begin{aligned}
\widehat{\psi}_0^{mle} &= 0.9506173 \\
L\left(\widehat{\psi}_0^{mle}\right) &= L(0.9506173) \\
&= \binom{170}{8}(\frac{1-0.9506173}{2-0.9506173})^8(1-\frac{1-0.9506173}{2-0.9506173})^{162} \\
&= 0.142988121 \\
\psi_0^{null} &= 0.3 \\
L\left(\psi_0^{null}\right) &= L(0.3) \\
&= \binom{170}{8}(\frac{1-0.3}{2-0.3})^8(1-\frac{1-0.3}{2-0.3})^{162} \\
&= 5.62253317 \times 10^{-28}
\end{aligned}
$$

We use the values above to calculate the likelihood ratio statistic $\Lambda$.

$$
\begin{aligned}
\Lambda &= \frac{L\left(\widehat{\psi}_0^{mle}\right)}{L\left(\psi_0^{null}\right)} \\
&= \frac{0.142988121}{5.62253317 \times 10^{-28}} \\
&= 2.543126 \times 10^{-26} \\
W &= 2\ln\Lambda \\
&= 2\ln\left(2.543126 \times 10^{-26}\right) \\
&= 121.6012
\end{aligned}
$$

We now calculate the p-value $P(\chi_1^2 \geq W) = P(\chi_1^2 \geq 121.6012)$.

```
p_value <- pchisq(121.6012, df=1, lower.tail=FALSE)

p_value
```

```
## [1] 2.822313e-28
```

With a p-value of $2.822313 \times 10^{-28}$, we reject the null hypothesis $H_0 : \psi_0 = 0.3$ under an alpha level $\alpha = 0.01$, supporting our alternative hypothesis $H_1 : \psi_0 > 0.3$ that the vaccine efficacy is greater than 0.3.

An alternative to using the $\chi^2_1$ distribution to calculate the p-value is to calculate the empirical p-value by generating a large number of samples from the null hypothesis. We then determine the fraction of samples with a likelihood ratio test statistic greater than $W$ to calculate the p-value.

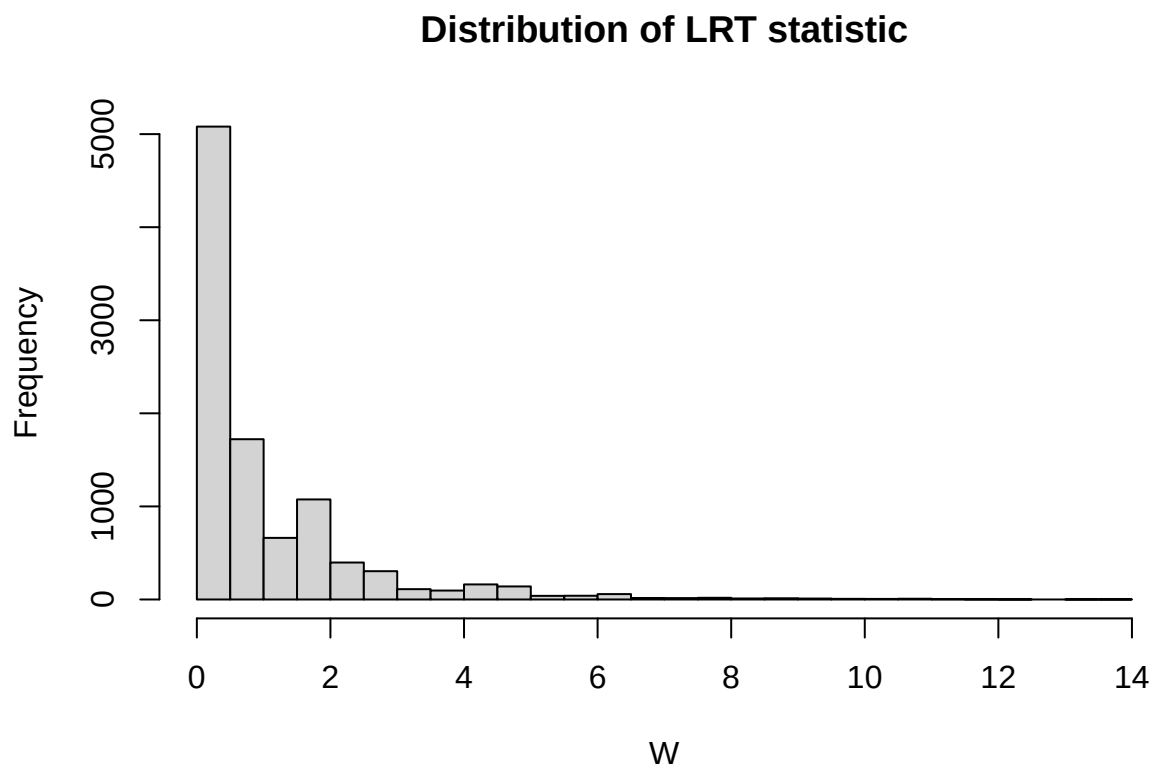
```r
set.seed(414)
n <- 170
x <- 8
psi_null <- 0.3
pi_null <- (1 - psi_null) / (2 - psi_null)
pi_mle <- summary(ml)$estimate[1]
W_obs <- 2 * ((log(choose(n, x)) + x * log(pi_mle) + (n - x) * log(1 - pi_mle)) -
              (log(choose(n, x)) + x * log(pi_null) + (n - x) * log(1 - pi_null)))

B <- 10000
w_star <- numeric(B)

for (i in 1:B) {
  x_star <- rbinom(1, size = n, prob = pi_null)
  pi_mle_star <- x_star/n

  w_star[i] <- 2 * ((log(choose(n, x_star)) +
                       x_star * log(pi_mle_star) +
                       (n - x_star) * log(1 - pi_mle_star)) -
              (log(choose(n, x_star)) +
                 x_star * log(pi_null) + (n - x_star) * log(1 - pi_null)))
}
```

```r
hist(w_star, breaks = 20, main = "Distribution of LRT statistic", xlab = expression(W))
```




```r
p_value_empirical <- sum(w_star > W_obs)/B
p_value_empirical
```

```
## [1] 0
```

With an empirical p-value of 0, we reach the same conclusion as we did with the p_value we reached assuming the likelihood ration test statistic $W$ follows a $\chi^2_1$ distribution. We reject the null hypothesis $H_0 : \psi_0 = 0.3$ under an alpha level $\alpha = 0.01$, supporting our alternative hypothesis $H_1 : \psi_0 > 0.3$ that the vaccine efficacy is greater than 0.3.

### Bayesian Inference

Detail the Bayesian approach.

From this, we can see that we would have an a value of 21.18 and a b value of 13.72. This would give us the following models:

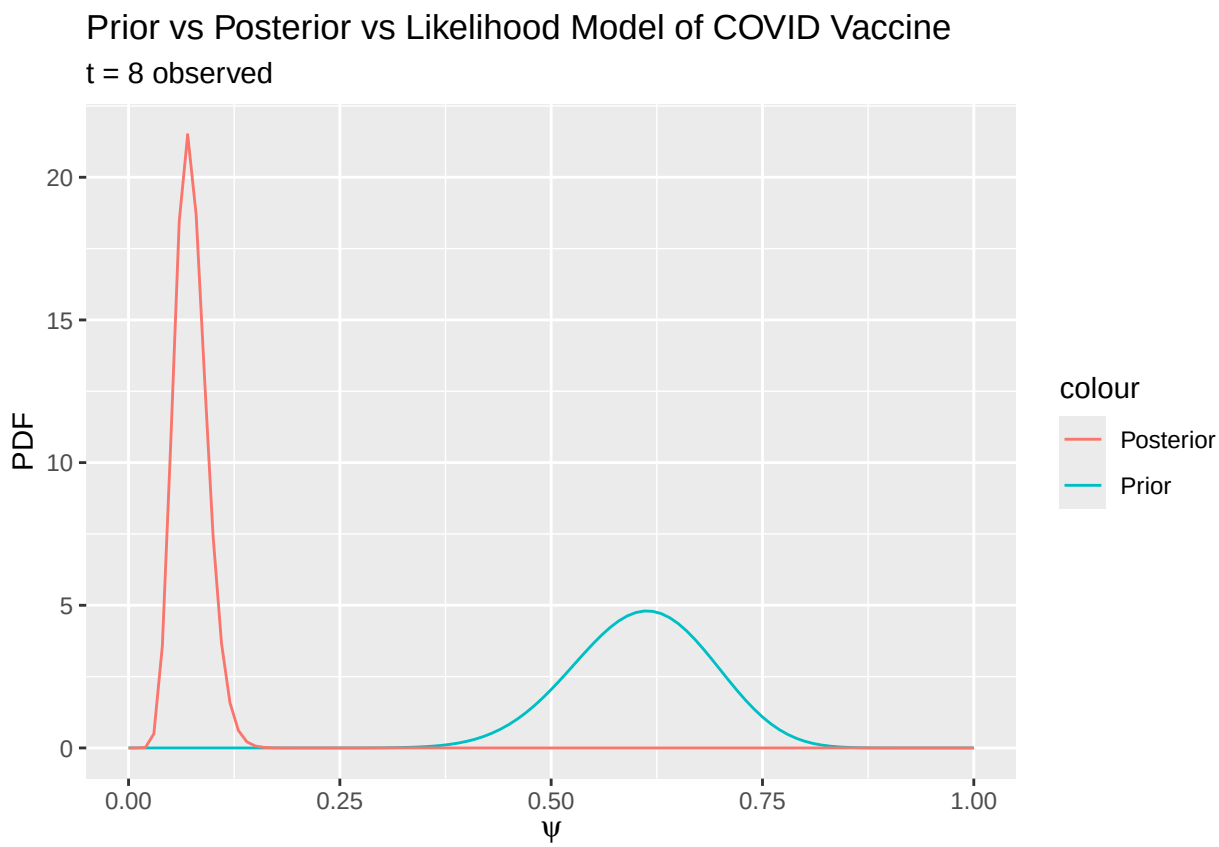
$$f(T|\pi) \sim Binom(n = 170, \pi) \text{ Data Model}$$

$$g(\pi) = Beta(a = 21.18, b = 13.72) \text{ Prior Model}$$

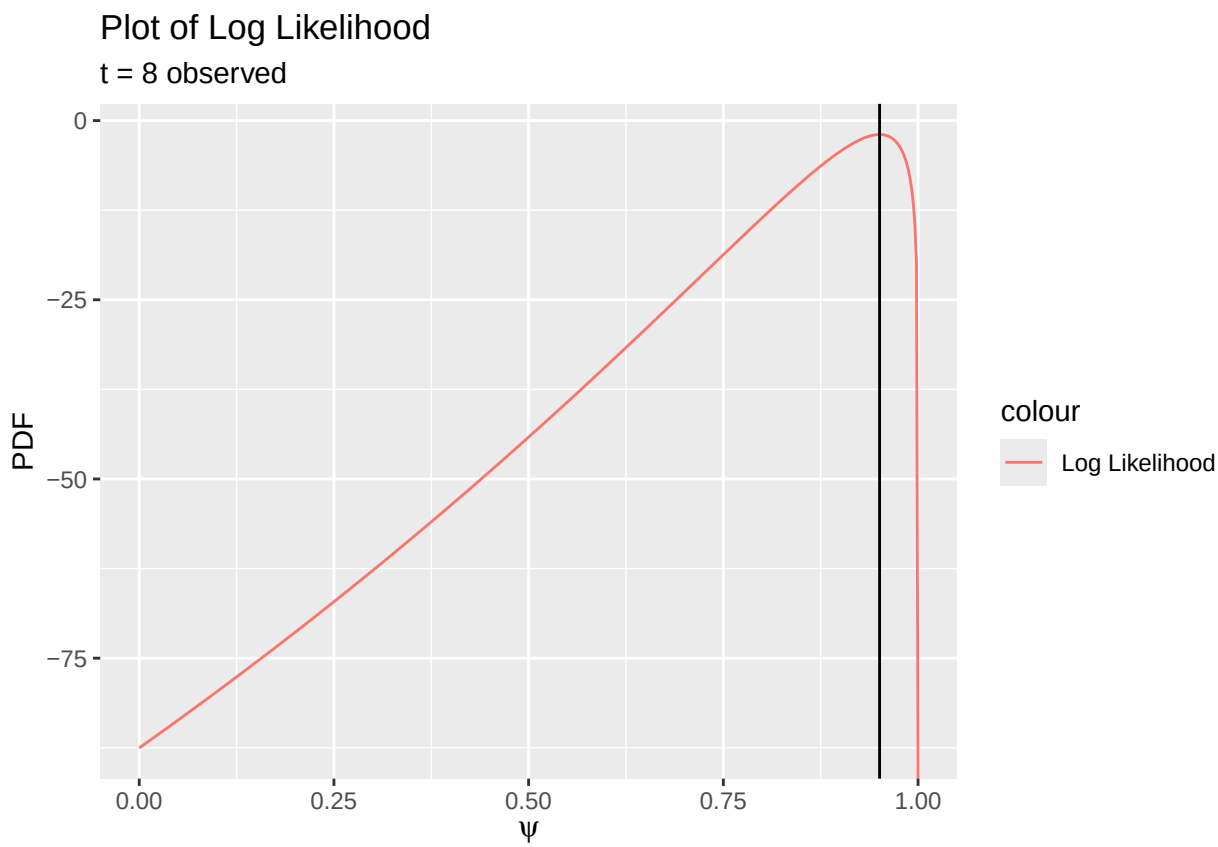
$$h(\pi|t) = Beta(a = 21.18 - t, b = 170 - t + 13.72)$$

```
loglik_binom <- function(psi, n, x){
  log(choose(n, x)) + x * log(1 - psi) - x * log(2 - psi) - (n - x) * log(2 - psi)
}


ggplot() +
  geom_function(fun = dbeta,
                mapping = aes(color = "Prior"),
                args = list(shape1 = ab[1], shape2 = ab[2]),
                xlim = c(0, 1)) +
  geom_function(fun = dbeta,
                mapping = aes(color = "Posterior"),
                args = list(shape1 = ab[1] - 8 + 1, shape2 = 170 - 8 + 1 + ab[2]),
                xlim = c(0, 1)) +
  labs(title = "Prior vs Posterior vs Likelihood Model of COVID Vaccine",
       subtitle = "t = 8 observed",
       x = expression(psi),
       y = "PDF")
```

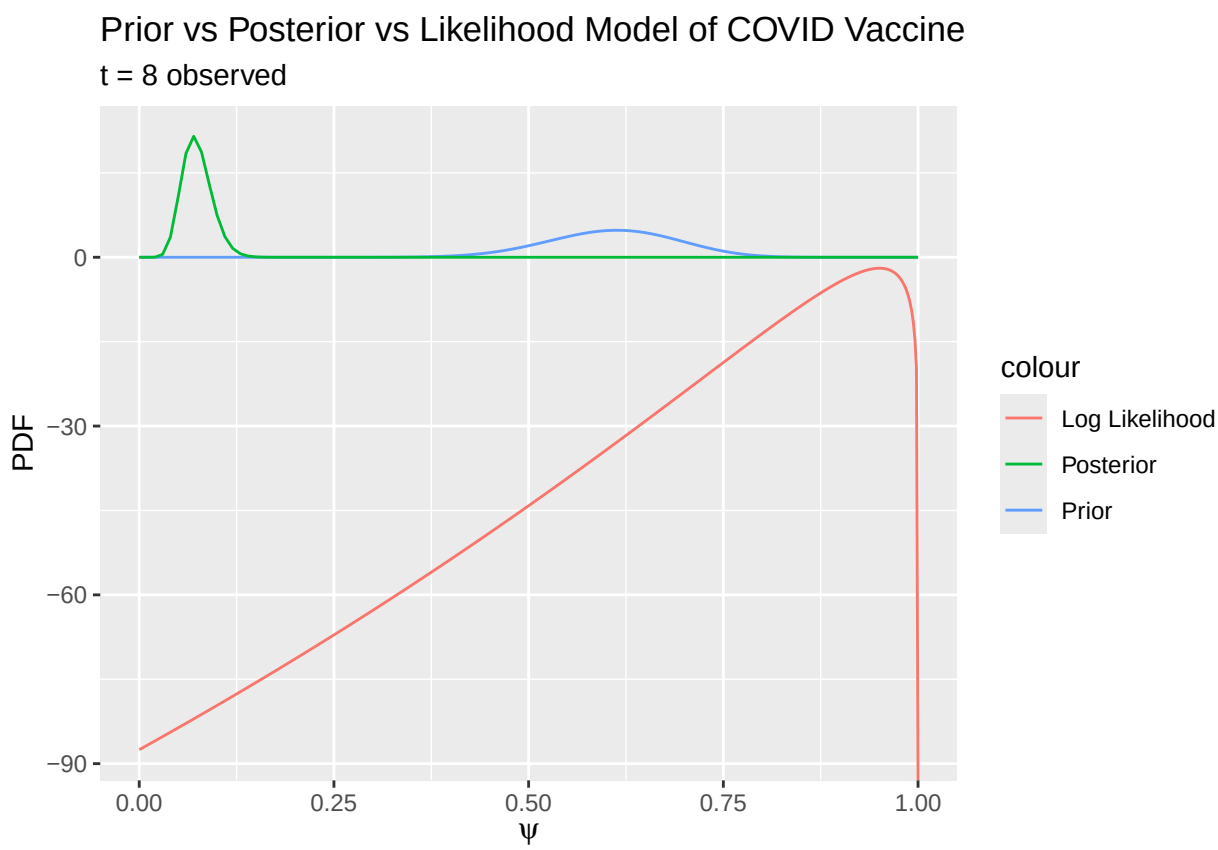



```r
ggplot() +
  geom_function(fun = loglik_binom,
                mapping = aes(color = "Log Likelihood"),
                args = list(n = 170, x = 8),
                xlim = c(0, 1),
                n = 500) +
  geom_vline(xintercept = 0.9506173) +
  labs(title = "Plot of Log Likelihood",
       subtitle = "t = 8 observed",
       x = expression(psi),
       y = "PDF")
```




```
ggplot() + # Icky for some reason, but I think this is what we were supposed to do?
  geom_function(fun = dbeta, # I will look at this again later and see if I can find the error
                mapping = aes(color = "Prior"),
                args = list(shape1 = ab[1], shape2 = ab[2]),
                xlim = c(0, 1)) +
  geom_function(fun = dbeta,
                mapping = aes(color = "Posterior"),
                args = list(shape1 = ab[1] - 8 + 1, shape2 = 170 - 8 + 1 + ab[2]),
                xlim = c(0, 1)) +
  geom_function(fun = loglik_binom,
                mapping = aes(color = "Log Likelihood"),
                args = list(n = 170, x = 8),
                xlim = c(0, 1),
                n = 500) +
```

```r
labs(title = "Prior vs Posterior vs Likelihood Model of COVID Vaccine",
     subtitle = "t = 8 observed",
     x = expression(psi),
     y = "PDF")
```




## Results

Present your findings.

## Discussion / Conclusion

Discuss / conclude here.

# Bibliography


Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from https://www.example.com.


# Appendix

## Code

```
# You can reference your code in the appendix (sample here).
df <- data.frame("Vaccine" = c(rep("BNT162b2", times = 8 + 17411),
                               rep("placebo", times = 162 + 17511)),
                 "covid" = c(rep("yes", times = 8),
                             rep("no", times = 17411),
                             rep("yes", times = 162),
                             rep("no", times = 17511)))
df %>% tabyl(Vaccine, covid) %>% adorn_percentages()%>% adorn_totals %>% adorn_ns()


mosaicplot(table(df))
```

## Proofs

### Proof A

Let $X$ denote the number of COVID-19 cases among the $n_1$ patients randomly assigned to the vaccine group and $Y$ denote the number of COVID-19 cases among the $n_2$ patients randomly assigned to the placebo group. Restricting our attention to the total of 170 COVID-19 cases

observed in the study, we want to know the PMF of the random variable $T$ that is defined as the count in the vaccine group given this restriction, or $T = X|X + Y = 170$.

Since the sample sizes $n_1$ and $n_2$ are large, and the rate that an individual contracts COVID-19 is small, we can assume

$$X \approx Pois(n_1\pi_v) \quad \text{and} \quad Y \approx Pois(n_2\pi_p)$$

where $X$ and $Y$ are independent, since the Poisson distribution approximates a binomial with a large $n$ and small $p$ well, as in this scenario.

Therefore the PMF of $T$ can be found as follows:

$$\begin{aligned}
P(T = t) &= P(X = t|X_Y = n) && t = 0, 1, \ldots, n \\
&= \frac{P(X = t \cap X + Y = n)}{P(X + Y = n)} \\
&= \frac{P(X = t \cap Y = n - t)}{P(X + Y = n)} \\
&= \frac{P(X = t) \cdot P(Y = n - t)}{P(X + Y = n)} && \text{Since } X \text{ and } Y \text{ are independent}
\end{aligned}$$